

Ventral Striatal Dopamine Increases following Hippocampal Sharp-Wave Ripples

Authors: Miriam A. Janssen¹, Hung-tu Chen¹, Nicolas X Tritsch^{2,3}, Matthijs A. A. van der Meer^{1*}

¹Department of Psychological and Brain Sciences, Dartmouth College, Hanover NH 03755. ²Department of Psychiatry, Douglas Hospital Research Centre, McGill University, Montreal, Quebec, Canada. ³Neuroscience Institute, New York University Grossman School of Medicine, New York, NY, USA. *Correspondence should be addressed to MvdM, Department of Psychological and Brain Sciences, Dartmouth College, 3 Maynard St, Hanover, NH 03755. E-mail: mvdm@dartmouth.edu.

Funding: This work was supported by the National Institute of Mental Health grant NIH R01 MH123466 (to MvdM) and the National Science Foundation NSF GRFP 2236868 (to MJ).

Conflict of Interest: The authors declare no competing financial interests.

Acknowledgements: We would like to thank the Neural Systems and Behavior Summer School at Woods Hole for the fruitful discussion and experimental pilot for this research.

25

30

35

Abstract

Leading theories suggest that hippocampal replay drives offline learning through coupling with an internal teaching signal such as ventral striatal dopamine (DA); however, the relationship between hippocampal replay and dopamine is unknown. Simultaneous recording of putative hippocampal replay events (dorsal CA1 sharp-wave ripples, SWRs) and fiber photometry of ventral striatal DA in mice revealed a significant increase in DA following SWRs, peaking ~0.3 s after SWRs. This result establishes for the first time a core theoretical requirement of offline learning in the mammalian brain.

Keywords: replay, reward prediction error, sharp wave ripples, dopamine, fiber photometry, hippocampus, ventral striatum

Introduction

The sequential reactivation of task-relevant activity in the hippocampus, colloquially referred to as replay, is thought to contribute to behavior through both online and offline processes. During sleep and awake rest, replay supports memory consolidation, broadly defined as the stabilization and updating of knowledge structures within and outside the hippocampus for later use. On-line, during active task engagement, replay may also support decision making through episodic memory retrieval and the generation of potential future scenarios (Foster, 2017; Joo & Frank, 2018; Olafsdottir et al., 2017; Pfeiffer, 2020), but see (van der Meer & Bendor, 2025). While the precise benefits of hippocampal replay are a subject of active debate, numerous studies have observed behavioral impairments when sharp wave-ripples (SWRs; the network-level signature of putative replay events) are disrupted (awake: (Jadhav et al., 2012), sleep: (Ego-Stengel & Wilson, 2010; Girardeau et al., 2009; Gridchyn et al. 2020); and behavioral enhancements when SWRs are augmented (de Lavilléon et al., 2015; Fernández-Ruiz et al., 2019), demonstrating that SWRs and replay are not mere epiphenomena. However, the neurophysiological mechanisms that enable replay to contribute to behavioral performance, whether for immediate or later use, remain unknown.

Theories of how replay influences immediate or future behavior consistently highlight the need for an evaluative signal to follow replay events. Intuitively, the online episodic retrieval of a relevant memory or the generation of a possible course of action should contribute only to taking that course of action if it is judged to lead to a positive outcome (Daw & Dayan, 2014; Lengyel & Dayan, 2007; Pezzulo et al., 2019; Redish, 2016). Similarly, a leading reinforcement learning account of memory consolidation posits that replay implements fictive learning driven by replayed experience, relying on a teaching signal to update value estimates (Matar & Daw, 2018; Momennejad et al., 2018; Sagiv et al., 2024). The brain signal most consistently associated with evaluation and teaching signals is the neuromodulator dopamine (DA), which reflects multiple reward- and reward prediction error-related signals that are especially prominent in the ventral striatum (Lerner et al., 2021). Interestingly, hippocampal activity is known to affect ventral striatal DA: for instance, stimulation of the ventral subiculum in particular causes DA release (Blaha et al., 1997), and hippocampal lesion or inactivation affects the activity of midbrain DA neurons (Lodge & Grace, 2007; Zhang et al., 2024). While one previous study found a relationship between hippocampal SWRs and the activity of reward-related VTA neurons (Gomperts et al. 2015), it is not known whether any coupling between hippocampal SWRs and ventral striatal dopamine exists.

To investigate this relationship, we recorded local field potentials in the dorsal CA1 of freely moving mice to detect sharp-wave ripples (SWRs) as markers of putative replay events, while simultaneously measuring DA concentration ([DA]) in the ventral striatum using fiber photometry of the GRAB_{DA2m} sensor (Sun et al., 2020). We hypothesized that hippocampal replay is coupled with teaching signals expressed as transient changes in ventral striatal [DA]. To test this hypothesis, we detected SWRs and measured [DA] while wild-type C57BL6/J mice ($n = 8$; 5 male, 3 female) rested before and after a reward-approach task designed to reveal putative reward prediction errors.

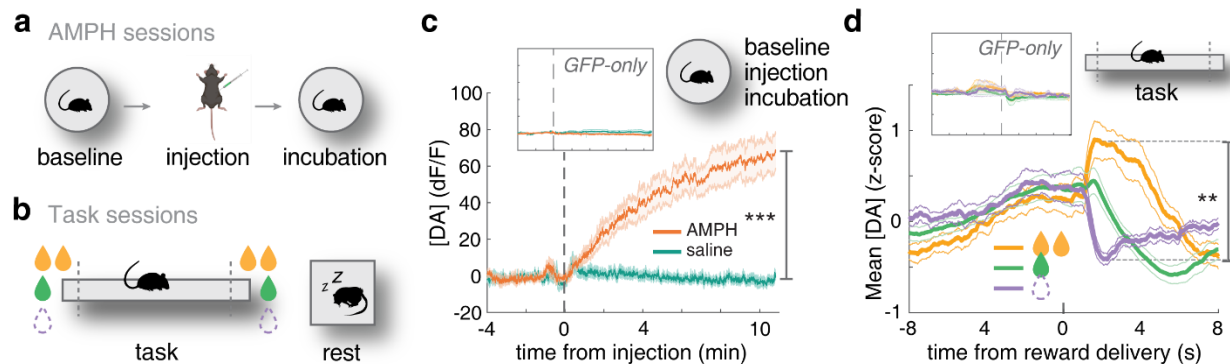


Figure 1 | Ventral striatal DA tracks reward outcomes. **a**, Schematic of the experimental design for amphetamine injection sessions, showing a baseline rest period, intraperitoneal injection of amphetamine or saline, and an incubation rest period. **b**, Schematic of the experimental design for task sessions, consisting of a probabilistic reward approach task on a linear track and rest periods. The *rest* epochs occur before and after track epochs to test if DA is coupled with SWRs during offline rest. The *task* epoch tests whether [DA] scales bidirectionally with reward: a positive putative prediction error occurs when animals receive 2x volume of evaporated milk reward at either end of the track (10 percent of trials; orange), no prediction error when animals receive expected 1x volume of reward (80 percent of trials; green), and a negative prediction error when animals receive no reward (10 percent of trials; purple). **c**, average [DA] fiber trace (dF/F) following amphetamine (AMPH, 0.625 – 2.5 mg/kg, i.p.) or saline injection. Change in average DA was measured as peak DA within 10 minutes of injection time subtracted by instantaneous DA at injection time (two sample t-test, $t(14) = -5.34$, $p = 1.05 \times 10^{-4}$). Inset: average [DA] fiber trace (dF/F) following AMPH (2.5 mg/kg) in control mice expressing GFP only (two sample t-test, $t(4) = -2.20$, $p = 0.097$). Shaded region indicates $\pm$ SEM ($n = 8$ mice for GRAB_{DA2m}, $n = 3$ for GFP-only). **d**, Putative reward prediction error signal in the average [DA] fiber trace (z-scored) aligned to reward zone entry, separated by reward magnitude: unexpected high reward (orange), expected medium sized reward (green), or unexpected omission of reward (purple). (paired, one-tailed Wilcoxon Signed Rank test between omission and high-volume trials, $p = 0.0039$). Inset: absence of RPE signal in GFP-only mice. (paired, one-tailed Wilcoxon Signed Rank test between omission and high-volume trials, $p = 0.125$). Shaded region indicates $\pm$ SEM ($n = 8$ mice for GRAB_{DA2m}, $n = 3$ for GFP-only). ** indicates $p < 0.01$, *** indicates $p < 0.001$.

Validating Dopamine Photometry with Amphetamine and Reward

First, we sought to validate the fluorescent dopamine sensor GRAB_{DA2m} (Sun et al., 2020) and to calibrate [DA] changes relative to *bona fide* reward prediction errors evoked by the unexpected omission or delivery of rewards. To confirm that the fluorescence signal measured reflects [DA], we first systemically administered amphetamine (AMPH), a dopamine reuptake inhibitor (Figure 1a). As expected, AMPH evoked a clear and persistent increase in ventral striatal [DA] compared to saline controls (peak values in the 11-minute post-injection period subtracted by instantaneous [DA] at time of injection between AMPH and control sessions, two sample t-test, $t(14) = -5.34$, $p = 1.05 \times 10^{-4}$, Figure 1c and Extended Data Figure 4). We ran this same validation step on three mice expressing a green fluorescent protein insensitive to [DA] (“GFP-only” mice) to control for non-specific changes in fluorescence linked to changes in breathing, movement or hemodynamics. As expected, GFP-only peak values in the 11-minute post-injection incubation period were not significantly different between AMPH and control sessions (two sample t-test, two sample t-test, $t(4) = -2.20$, $p = 0.097$; Figure 1c, inset).

Next, to obtain putative reward prediction error (RPE) signals during behavior, we trained mice to perform a probabilistic reward delivery task, where rewards of different sizes were delivered at both ends of a linear track: medium-sized rewards (evaporated milk) on 80% of trials, large rewards (2x the medium reward) on 10% of trials, and no reward at all on the remaining 10% of trials (Figure 1b). Consistent with previous work, we found a clear suppression of the [DA] signal following reward omission relative to the expected (medium) amount, and a clear increase in [DA] following larger-than-expected reward; Figure 1d). Specifically, the high-volume reward trials had a mean peak [DA] fiber value of 1.11 ± 0.15 SEM, the medium volume reward trials had a mean peak [DA] fiber value of 0.64 ± 0.15 SEM, and the omission trials had a mean trough [DA] fiber value of -0.48 ± 0.07 SEM (paired, one-tailed Wilcoxon Signed Rank test between omission and high-volume trials, $p = 0.0039$). GFP-only mice did not show any significant difference between the peak and trough averaged [DA] values between high volume and omission trials (paired, one-tailed Wilcoxon Signed Rank test between omission and high-volume trials, $p = 0.125$, Figure 1d, inset). Thus, consistent with previous results, both pharmacological and behavioral validation of GRAB_{DA2m} demonstrates that we are measuring a putative [DA] RPE signal from the ventral striatum.

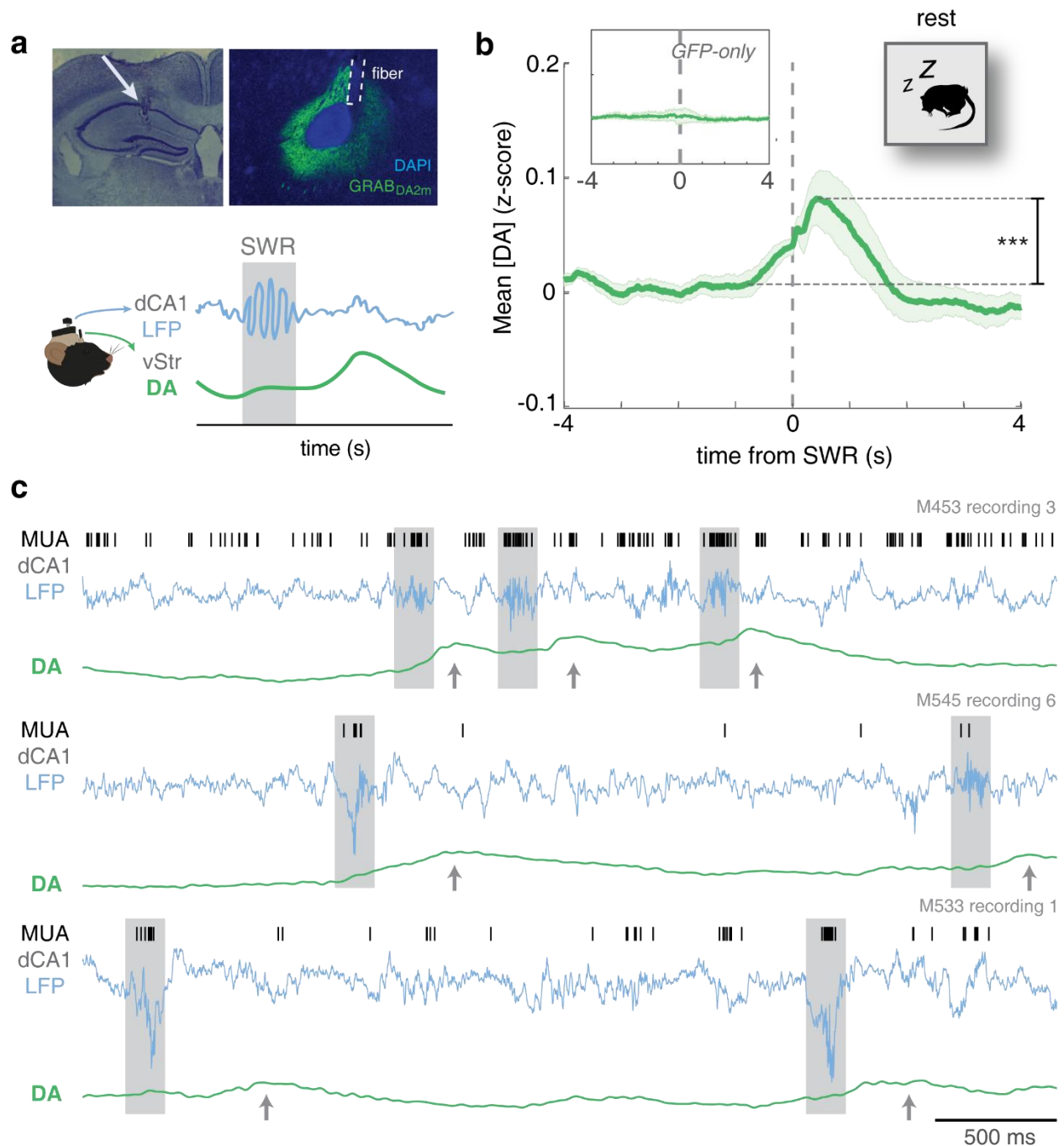


Figure 2 | Ventral striatal dopamine increases following hippocampal sharp wave-ripples. **a**, Schematic of unilateral (5 right, 3 left) microelectrode and fiber photometry implantation along with example coronal sections of dorsal CA1 gliosis indicating microelectrode placement and ventral striatal GRAB_{DA2m} expression for fiber photometry (DAPI, blue; GFP, green). For more see Extended Data Figure 1. White dashed lines indicate fiber placement. Schematic of hypothesized results, where the ventral striatal fiber photometry dopamine signal (DA, green line) increases following dorsal CA1 sharp wave-ripples (SWR, blue line and shaded region). On average, we expect to see an increase in fiber photometry signal (average

170

[DA], green line) following SWRs (SWR onset, dashed gray line). **b**, average ventral striatal [DA] increase following SWRs (dashed vertical line at $t = 0$). Inset: as in **b**, for GFP only mice. Shaded region indicates $\pm$ SEM ($n = 8$ mice for GRAB-DA, $n = 3$ mice for GFP-only). **c**, Example multiunit activity (MUA, black tick marks), local field potential traces (LFP, blue line) and GRAB_{DA2m} fiber photometry traces (detrended and z-scored DA signal, green line) for three sessions. Detected SWRs as seen by the high-frequency local field potential oscillations (M453, M545, M533) with lower frequency deflections (M533) and corresponding MUA are shaded in grey. Following these detected SWRs, there are transients in the DA signal indicated by the grey arrows.

Ventral Striatal Dopamine Increases Following Hippocampal Sharp-Wave Ripples

Having established we could measure ventral striatal dopamine reflecting putative reward prediction errors, we next examined the relationship between hippocampal SWRs and ventral striatum [DA] during rest periods (Figure 2a). To determine statistically if [DA] increased after SWRs rather than before SWRs, we ran session-based linear mixed effects regression (LMM) with session averaged peak [DA] values in the one second preceding or following a SWR as the dependent variable. This model was improved by labeling these peak values as before or after a SWR (likelihood ratio test, $LR_{Stat_{\Delta df=1}} = 16.87$, $p = 9.99 \times 10^{-4}$) and the peak DA value was significantly larger after compared to before a SWR ($\beta = 0.08$, $t(182) = 4.21$, $p = 3.96 \times 10^{-5}$, $CI: 0.07, 0.15$) (Figure 2b). There was no before-after difference in the GFP-only mice (likelihood ratio test, $LR_{Stat_{\Delta df=1}} = 0.0023$, $p = 0.96$). To visualize this SWR aligned peak, for each daily recording session, we generated SWR-triggered [DA] (SWR-[DA]) peri-event time histograms (PETHs). Across all mice and sessions, we observed a clear increase in [DA] following SWRs (Figure 2b; magnitude: 0.11 ± 0.018 SEM z-score; delay: 0.35 ± 0.15 SEM seconds after detected SWRs). To contextualize this result, we compared the size of the internally generated SWR-DA transients with the DA transients generated following the receipt of larger-than-expected rewards (putative positive RPEs). The SWR-DA transient was therefore 23.55% of the size of a putative positive RPE, which was determined by subtracting the size of the [DA] peak transient from the expected medium volume trials from the high-volume trials.

To test if this SWR-triggered DA increase is experience-dependent, we compared SWR-triggered [DA] PETHs between pre- and post-task rest epochs. During pre-task rest epochs, peak [DA] values imaged within one second of a SWR averaged 0.11 ± 0.02 SEM z-score (delay: 0.37 ± 0.17 SEM, and the average post-task rest session fiber peak value was 0.12 ± 0.02 SEM z-score occurring at 0.38 ± 0.15 SEM seconds. We found that a session-based LMM with peak fiber value

z-score within one second of after an SWR as the dependent variable was not improved by indicating if the session was a pre- or post-task rest session (likelihood ratio test, $LRStat_{\Delta df=1} = 0.94$, $p = 0.33$) (Extended Data Figure 5a). Similarly, an LMM was not improved by the addition of if averaged SWR-DA peaks were from an early (sessions 1-4) vs. late (sessions 5-8) session (likelihood ratio test, $LRStat_{\Delta df=1} = 0.04$, $p = 0.86$) (Extended Figure 5b). Thus, we found no evidence that SWR-DA coupling is increased following experience on the track.

By inspecting individual sessions, we observed notable variability in the magnitude and timing of SWR-DA PETHs (Figure 2c, Extended Data Figure 6). To describe this variability, we computed single-session PETHs, which we compared to a distribution of resampled PETHs in which the [DA] signal was circularly shifted by a random amount. To determine SWR-DA strength for each session, we z-scored the average [DA] trace relative to the resampled session-wide mean and standard deviation. If the observed session [DA] trace is greater or less than 1.96 z-score relative to the resampled data (corresponding to $p < 0.05$), that is taken as evidence of a significant SWR-[DA] association. Out of 92 sessions, 10 sessions had a positive association, and none had a negative association. 3 of those sessions were pre-task rest sessions and 7 were post-task rest sessions. This variability across sessions could not be explained by differences in running speed on the track (a proxy for motivation; likelihood ratio test, $LRStat_{\Delta df=1} = 0.28$, $p = 0.62$) or by the magnitude of reward prediction error coding during the task (as quantified by the t-value between the peak [DA] signal response for high volume trials and the trough for omission trials, likelihood ratio test, $LRStat_{\Delta df=1} = 0.39$, $p = 0.55$) (Extended Data Figure 7). Thus, although the grand average across all mice and sessions shows a clear SWR-DA relationship, this coupling is variable across individual recording sessions.

In sum, this study establishes a link between sharp wave-ripples and transient dopamine increases in the ventral striatum, providing novel mechanistic insight into how putative memory traces may be evaluated and used to guide learning. This result speaks directly to influential theoretical accounts of replay-based offline reinforcement learning such as DYNA and its descendants (Matar & Daw, 2018; Sagiv et al., 2024; Sutton, 1990), which require that replay events generate a teaching signal that updates value estimates used for later decision-making (i.e. implement temporal credit assignment). Our data provides that requirement, showing that SWRs are followed by a dopamine transient of the kind necessary for credit assignment. By measuring dopamine release, our work advances earlier findings (Gomperts et al., 2015) by showing direct evidence that hippocampal replay is subject to dopaminergic evaluation in the

ventral striatum. Mechanistically, long-term potentiation at striatal synapses is strongly modulated by dopamine release, and the fast timescale of replay puts it in an ideal position to drive LTP (Buzsáki, 1989; Carr et al., 2011). Beyond reinforcement learning, our finding also bears on the traditional memory consolidation literature, where replay implements the stabilization, transfer and reorganization of memory traces (Squire et al., 2015). A long-standing challenge for these accounts is explaining how certain events provide lasting consequences, while others are forgotten; a common proposal is that replay is gated or tagged for consolidation based on some kind of salience signal (e.g. Rouhani et al., 2023; Yang et al., 2024). Our observation that on average, increases in DA tend to follow SWRs (by about 0.3 s) could provide such a signal, with dopamine acting as a modulator of SWR-induced plasticity that helps determine which events are integrated into long-term cortico-striatal memory structures.

While we observed reliable SWR-DA coupling on average, its session-to-session variability was not explained by motivation, task experience, or the strength of task-related RPEs, highlighting the need for future work linking the content of replay to the valence and magnitude of ensuing dopamine transients. Such work should establish whether replay of positive or negative outcomes differentially recruits dopaminergic teaching signals, bridging cognitive models of replay-based learning with their mechanistic underpinnings. Finally, the circuit mechanisms underlying this SWR-DA coupling remain an open question. One possibility is that the hippocampus modulates dopamine release in the ventral striatum through direct projections to presynaptic DA terminals (Blaha et al., 1997). Alternatively, or additionally, hippocampal output via the subiculum may influence VTA activity through polysynaptic loops involving the nucleus accumbens and ventral pallidum (Lisman & Grace, 2005). Disentangling these possibilities will require causal manipulations and projection-specific targeting, but either pathway would support the broader conclusion: that hippocampal replay events have the potential to be coordinated with evaluative systems in the brain, shaping both the content of memory and the structure of future decisions.

References

- Blaha, C. D., Yang, C. R., Floresco, S. B., Barr, A. M., & Phillips, A. G. (1997). Stimulation of the Ventral
275 Subiculum of the Hippocampus Evokes Glutamate Receptor-mediated Changes in Dopamine
Efflux in the Rat Nucleus Accumbens. *European Journal of Neuroscience*, 9(5), 902–911.
<https://doi.org/10.1111/j.1460-9568.1997.tb01441.x>
- Buzsáki, G. (1989). Two-stage model of memory trace formation: A role for “noisy” brain states.
Neuroscience, 31(3), 551–570. [https://doi.org/10.1016/0306-4522\(89\)90423-5](https://doi.org/10.1016/0306-4522(89)90423-5)
- 280 Carey, A. A., Tanaka, Y., & van der Meer, M. A. A. (2019). Reward revaluation biases hippocampal replay
content away from the preferred outcome. *Nature Neuroscience*, 22(9), 1450–1459.
<https://doi.org/10.1038/s41593-019-0464-6>
- Carr, M. F., Jadhav, S. P., & Frank, L. M. (2011). Hippocampal replay in the awake state: A potential
physiological substrate of memory consolidation and retrieval. *Nature Neuroscience*, 14(2), 147–
285 153. <https://doi.org/10.1038/nn.2732>
- Daw, N. D., & Dayan, P. (2014). The algorithmic anatomy of model-based evaluation. *Philosophical
Transactions of the Royal Society B: Biological Sciences*, 369(1655), 20130478.
<https://doi.org/10.1098/rstb.2013.0478>
- de Lavilléon, G., Lacroix, M. M., Rondi-Reig, L., & Benchenane, K. (2015). Explicit memory creation
290 during sleep demonstrates a causal role of place cells in navigation. *Nature Neuroscience*, 18(4),
493–495. <https://doi.org/10.1038/nn.3970>
- Ego-Stengel, V., & Wilson, M. A. (2010). Disruption of ripple-associated hippocampal activity during rest
impairs spatial learning in the rat. *Hippocampus*, 20(1), 1–10. <https://doi.org/10.1002/hipo.20707>
- Fernández-Ruiz, A., Oliva, A., Fermino De Oliveira, E., Rocha-Almeida, F., Tingley, D., & Buzsáki, G.
295 (2019). Long-duration hippocampal sharp wave ripples improve memory. *Science*, 364(6445),
1082–1086. <https://doi.org/10.1126/science.aax0758>
- Foster, D. J. (2017). Replay Comes of Age. *Annual Review of Neuroscience*, 40(1), 581–602.
<https://doi.org/10.1146/annurev-neuro-072116-031538>

- Girardeau, G., Benchenane, K., Wiener, S. I., Buzsáki, G., & Zugaro, M. B. (2009). Selective suppression
300 of hippocampal ripples impairs spatial memory. *Nature Neuroscience*, 12(10), 1222–1223.
<https://doi.org/10.1038/nn.2384>
- Gomperts, S. N., Kloosterman, F., & Wilson, M. A. (2015). VTA neurons coordinate with the hippocampal
reactivation of spatial experience. *eLife*, 4(OCTOBER2015).
<https://doi.org/10.7554/eLife.05360.001>
- 305 Jadhav, S. P., Kemere, C., German, P. W., & Frank, L. M. (2012). Awake hippocampal sharp-wave
ripples support spatial memory. *Science*, 336(6087), 1454–1458.
<https://doi.org/10.1126/science.1217230>
- Joo, H. R., & Frank, L. M. (2018). The hippocampal sharp wave–ripple in memory retrieval for immediate
use and consolidation. *Nature Reviews Neuroscience*, 19(12), 744–757.
- 310 Lengyel, M., & Dayan, P. (2007). Hippocampal Contributions to Control: The Third Way. *Advances in
Neural Information Processing Systems*, 20.
[https://proceedings.neurips.cc/paper/2007/hash/1f4477bad7af3616c1f933a02bfabe4e-
Abstract.html](https://proceedings.neurips.cc/paper/2007/hash/1f4477bad7af3616c1f933a02bfabe4e-Abstract.html)
- Lerner, T. N., Holloway, A. L., & Seiler, J. L. (2021). Dopamine, Updated: Reward Prediction Error and
315 Beyond. *Current Opinion in Neurobiology*, 67, 123–130.
<https://doi.org/10.1016/j.conb.2020.10.012>
- Lisman, J. E., & Grace, A. A. (2005). The Hippocampal-VTA Loop: Controlling the Entry of Information
into Long-Term Memory. *Neuron*, 46(5), 703–713.
- Lodge, D. J., & Grace, A. A. (2007). Aberrant Hippocampal Activity Underlies the Dopamine
320 Dysregulation in an Animal Model of Schizophrenia. *The Journal of Neuroscience*, 27(42),
11424–11430. <https://doi.org/10.1523/JNEUROSCI.2847-07.2007>
- Mattar, M. G., & Daw, N. D. (2018). Prioritized memory access explains planning and hippocampal replay.
Nature Neuroscience, 21(11), 1609–1617. <https://doi.org/10.1038/s41593-018-0232-z>
- Momennejad, I., Otto, A. R., Daw, N. D., & Norman, K. A. (2018). Offline replay supports planning in
325 human reinforcement learning. *eLife*, 7. <https://doi.org/10.7554/eLife.32548>

Olafsdottir, G., Cloke, P., & Vögele, C. (2017). Place, green exercise and stress: An exploration of lived experience and restorative effects. *Health & Place*, 46, 358–365.

<https://doi.org/10.1016/j.healthplace.2017.02.006>

Pezzulo, G., Donnarumma, F., Maisto, D., & Stoianov, I. (2019). Planning at decision time and in the background during spatial navigation. *Current Opinion in Behavioral Sciences*, 29, 69–76.

<https://doi.org/10.1016/j.cobeha.2019.04.009>

Pfeiffer, B. E. (2020). The content of hippocampal “replay.” *Hippocampus*, 30(1), 6–18.

<https://doi.org/10.1002/hipo.22824>

Redish, A. D. (2016). Vicarious trial and error. *Nature Reviews Neuroscience*, 17(3), 147–159.

<https://doi.org/10.1038/nrn.2015.30>

Rouhani, N., Niv, Y., Frank, M. J., & Schwabe, L. (2023). Multiple routes to enhanced memory for emotionally relevant events. *Trends in Cognitive Sciences*, 27(9), 867–882.

<https://doi.org/10.1016/j.tics.2023.06.006>

Sagiv, Y., Akam, T., Witten, I. B., & Daw, N. D. (2024). Prioritizing replay when future goals are unknown.

bioRxiv. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10942393/>

Squire, L. R., Genzel, L., Wixted, J. T., & Morris, R. G. (2015). Memory Consolidation. *Cold Spring Harbor Perspectives in Biology*, 7(8), a021766. <https://doi.org/10.1101/cshperspect.a021766>

Sun, F., Zhou, J., Dai, B., Qian, T., Zeng, J., Li, X., Zhuo, Y., Zhang, Y., Wang, Y., Qian, C., Tan, K., Feng, J., Dong, H., Lin, D., Cui, G., & Li, Y. (2020). Next-generation GRAB sensors for monitoring dopaminergic activity in vivo. *Nature Methods*, 17(11), Article 11. <https://doi.org/10.1038/s41592-020-00981-9>

Sutton, R. S. (1990). First results with Dyna, an integrated architecture for learning, planning and reacting. *Neural Networks for Control*, 179.

van der Meer, M. A. A., & Bendor, D. (2025). Awake replay: Off the clock but on the job. *Trends in Neurosciences*, 48(4), 257–267. <https://doi.org/10.1016/j.tins.2025.02.006>

Yang, W., Sun, C., Huszár, R., Hainmueller, T., Kiselev, K., & Buzsáki, G. (2024). Selection of experience for memory by hippocampal sharp wave ripples. *Science*, 383(6690), 1478–1483.

Zhang, Z., Takahashi, Y. K., Montesinos-Cartegena, M., Kahnt, T., Langdon, A. J., & Schoenbaum, G.

(2024). Expectancy-related changes in firing of dopamine neurons depend on hippocampus.

355 *Nature Communications*, 15(1), 8911. <https://doi.org/10.1038/s41467-024-53308-z>

Methods

Subjects and overall timeline. Data was collected from eight 5-10 months-old wild-type mice (C57BL/6J, 3 female, 5 male), ranging between 22–31 grams in weight. For the GFP-only study, data was collected from three 3-10 months-old wild-type mice (C57BL/6J, 3 male), approximately 24–33 grams in weight.

Briefly, mice underwent a single recovery surgery to receive (1) intracranial adeno-associated virus (AAV) injection for the expression of the dopamine sensor GRAB_{DA2m} (Sun et al., 2020), (2) an optic fiber implant to image [DA] in the ventral striatum, and (3) an electrode implant in the ipsilateral hippocampus (dorsal CA1) for the recording of sharp wave-ripples (SWRs). Following recovery from surgery, mice were food-restricted and pretrained in daily sessions to shuttle back and forth on a linear track for liquid rewards before data acquisition began. All procedures were approved by the Institutional Animal Care and Use Committee at Dartmouth College (Dartmouth College IACUC protocol #00002235).

Surgery. Before surgery, mice received a subcutaneous injection of 5 mg/kg ketoprofen and 4 mg/kg dexamethasone. Mice were anesthetized in a closed chamber with an isoflurane concentration of 5% vaporized in 2L/min medical-grade oxygen. Following induction, mice were transferred to the stereotaxic apparatus, where they continued to receive isoflurane through a nose cone at a concentration of 0.5–2% in 1 L/min oxygen. Mice were shaved to expose the skin, which was sanitized with betadine and ethanol, and anesthetized locally with lidocaine, before an incision was made. One metal screw was placed in the skull near the future location of the fiber implant to provide stability. A drill burr was used to make a craniotomy above the ventral striatum (from bregma, +1.4 mm anteroposterior and ± 0.9 –1.0 mediolateral). Mice received two injections of 200 nL AAV9-hsyn-GRAB_{DA2m} DA4.4 (WZ Biosciences, YL002009-AV9-PUB) (Sun et al., 2020) at 100 nL/min unilaterally in the ventral striatum (-3.8 and -4.1 mm dorsoventral from skull surface). GFP-only control mice were unilaterally injected with 200 nL AAV8-CamKII-Jaws-KGC-GFP (Addgene, 65015) at the same coordinates. After the viral infusion, a single 400 μ m optic fiber mated to a 1.25 mm ferrule was chronically implanted -3.8 mm below the skull surface at the ventral striatal injection site. The screws and ferrule were secured to the skull with dental cement (C&B Metabond, Parkell). A 1.8 mm trephine was then used to make a craniotomy centered above the dorsal CA1 (from bregma, -2.0 mm anteroposterior and ± 1.5 mediolateral). Stereotrode microdrives (Axona, described below) were lowered in the craniotomy so that the inner cannulae were implanted 0.2 mm below the pial surface. Two additional ground screws inserted above the ipsilateral cerebellum and contralateral posterior parietal cortex, connected to the microdrive and secured to the skull and microdrive with dental cement. After surgery, mice were injected subcutaneously with an antibiotic (enrosite ; 5 mg/kg) and saline (1 mL). 24h post-surgery, mice were injected subcutaneously with ketoprofen (5 mg/kg), enrosite (5 mg/kg), and saline (1mL); 48h post-surgery, mice were injected with 5 mg/kg of enrosite and 1 mL of saline.

Behavioral task. Following recovery from surgery (48h minimum), mice received two daily habituation sessions (lap training and exposure to the behavioral apparatus; 5 min each). Mice were then trained to criterion on the behavioral task (described below) until they ran 60 trials in a 30-minute session (training to criterion took 2–6 days). While mice were being trained to criterion, they were placed on food control, with access to approximately 2–3 grams of chow daily and *ad*

libitum water until the end of experiments. During food control, weights were maintained above 80% of baseline. In addition, electrode depths were adjusted to reach the hippocampal cell layer daily following behavioral training.

The behavioral task was run on a 1-meter-long linear track, where running from one end to the other triggers automated evaporated milk delivery. 10% of trials resulted in no reward, 80% in medium volume reward (~25 μ L), and 10% in large volume reward (~50 μ L). Dopamine (DA) fiber photometry signal screening began after at least 9 days of virus expression after surgery. Recordings started once [DA] transients and hippocampal SWRs could readily be observed, and once mice reached behavioral criterion on the task (i.e., 10-31 days post-surgery).

Each daily recording session consisted of a 30 min rest session in the sleep box (a 10 $\times$ 10 $\times$ 10-inch enclosure with bedding), a ~30 min task session on a linear track (rewarded, as outlined above), and a 30 min rest session in the sleep box. We aimed to record eight sessions for each mouse and four additional amphetamine validation sessions (described below). Infrared photobeam sensors detected entry into the reward zone where evaporated milk was dispensed. All data were acquired using a Neuralynx Digital Lynx recording system equipped with Hauppauge HVR USB2 video capture using a ceiling-mounted camera.

In some sessions, amphetamine was administered to validate the fluorescence signal was modulated by [DA]. Each of these sessions consisted of a 10-min baseline period preceding the injection of either amphetamine or saline (which alternated daily), followed by a 10 min incubation period. We administered either 0.625–2.5 mg/kg of amphetamine diluted in saline to a volume of 10% the weight of each mouse or just saline at a volume of 10% the weight of each mouse i.p. We adjusted the concentration to have no obvious motor effects while still recording SWRs.

Fiber photometry and microdrive implant construction. For hippocampal SWR recordings, we used 16-channel microdrives equipped with a MillMax connector (Axona Ltd). Prior to surgery, a single driveable bundle consisting of 8 stereotrodes (25 μ m nichrome wire diameter, California Fine Wire Co. RO-800) was cut in a staggered manner, approximately 0.5–1.5 mm in length from the microdrive inner cannula, to provide a depth profile across the hippocampal cell layer for recording SWRs. For fiber photometry, we used Neurophotometrics 0.37 NA black ceramic fibers, fiber diameter 400 μ m, ferrule diameter 1.25 mm. The fibers were cut to 6.5 mm in length, tested to allow 70% or better light transmission. At the start of each session, implanted microdrives were connected to the data acquisition system using a single Neuralynx HS-18-MM-LED headstage run through a motorized commutator, and signal quality (existence of hippocampal SWRs) checked visually before recording began.

Fiber photometry data acquisition and preprocessing. GRAB_{DA2m} fiber photometry was performed using a custom photometry system consisting of a photometry cube (FMC3_E(460-490)_F(500-550)_S, Doric) connected to (a) a 470 nm light-emitting diode (M470F3, Thorlabs) via a patch cord (MFP_600/630/-0.48_m_FCM-SMA, Doric), (b) the mouse via a fiber optic patch cable (MFP_400/460/1100-0.48_2m_FCM-MF1.25, Doric or M99L01, Thorlabs) and plastic coupling sleeve (1.25 mm, either from Neurophotometrics or Thorlabs), and (c) a photoreceiver (Model 2151, Newport) via a patch cord (MFP_600/630/-0.48_m_FCM-FCM, Doric). The photodetector output was fed into a DC-coupled analog input of the Neuralynx acquisition system

(Cheetah; 5,000 Hz sampling; 1,600 Hz sampling for GFP-only mice). At the start of each session, excitation light intensity was set to 50-100 μ W at the tip of the patch cord and the presence of [DA] transients was visually confirmed by streaming a copy of the photodetector output signal using WaveSurfer (Janelia) via a National Instruments ADC board.

For subsequent data analysis, the GRAB_{DA2m} signal was first downsampled to 1000 Hz, filtered using a 20 Hz lowpass Butterworth filter, linearly detrended using a 60-second moving window to account for gradual photobleaching (Chronux `locdetrend` function), and z-scored over the entire session. Signals from GFP-only mice were processed similarly, except that downsampling was omitted. For amphetamine and saline injection sessions the 60 s moving window detrending method was omitted due to the need to measure slow amphetamine-induced changes in the [DA] signal. Instead, we fit a double exponential to the first baseline session (pre-injection) and subtracted that from the signal recorded with amphetamine on board to control for photobleaching while retaining slower changes in [DA].

Simultaneous electrophysiology data acquisition and preprocessing. Local field potentials (LFPs) were acquired using a Neuralynx Digital Lynx data acquisition system (1500 Hz sampling; 1-500 Hz bandpass filter). Multi-unit activity was recorded by storing a 1-ms snapshot whenever a 32 kHz, 600–6000 Hz filtered signal on any stereotrode channel exceeded an experimenter-set threshold (typically 50 μ V). All signals were referenced against ground screws in ipsilateral cerebellum or contralateral posterior parietal cortex.

All analyses, described in detail below, were performed using MATLAB 2023a (MathWorks) and can be reproduced using code available through a public GitHub repository (<https://github.com/mimijanssen/replayDA>, using the van der Meer lab codebase: <https://github.com/vandermeerlab/vandermeerlab>).

Sharp wave-ripple event detection. SWR events were detected based on ripple-band LFP power using methods described in (Carey et al., 2019). For each recording session, the recording channel with the clearest SWRs was identified by the experimenter. On that channel, a noise-corrected SWR Fourier spectrum was obtained by manual identification of example SWR events and subtracting a non-SWR ‘noise’ Fourier spectrum (Carey et al. 2019). The dot product between this noise-corrected frequency spectrum and the frequency spectrum of the LFP was obtained using a 60 ms sliding window to obtain a SWR similarity score. Next, we computed a control similarity score by applying the same noise-corrected frequency spectrum dot product to a different recording channel that did not have SWRs; this control similarity score was subtracted from the SWR channel similarity score to obtain an “SWR difference score.” This SWR difference score was then z-scored across the entire session and thresholded at 3-5 standard deviations (SDs, depending on the session) above the mean to obtain candidate SWR events. The resulting candidates were further restricted to SWRs exceeding 20 ms in duration. Detected events that occurred within 20 ms of each other were merged as they were likely from the same SWR.

SWR-triggered (peri-event) DA calculations. For each session, we calculated the average [DA] trace surrounding SWR events (i.e., SWR-triggered DA average, or SWR-DA). To obtain the grand average trace shown in Figure 2a, we took the mean over sessions for each mouse and then took the mean over mice. For the single-session analyses described in the main text, we

compared the mean SWR-DA signal for a given session to the distribution of SWR-DA averages obtained from 1,000 resampled sessions in which the DA trace was circularly shifted by a random amount; this resampling procedure results in a permutation z-statistic which we use, first, to
 485 determine how many single sessions show significant SWR-DA coupling, and second, as a dependent variable in regressions (described below).

Session-based regression analyses. To assess whether DA transients occurred before or following SWRs, we ran session-based mixed-effects regressions of the following forms:

$$\text{SWRDA} \sim 1 + (1 \mid \text{Mouse}) + (1 \mid \text{Session})$$

490
$$\text{SWRDA} \sim 1 + \text{BeforeAfter} + (1 \mid \text{Mouse}) + (1 \mid \text{Session})$$

where the dependent variable *SWR-DA* denotes the peak of the session-averaged SWR-DA signal within one second (before or after) a SWR. Thus, each session contributes two data points: one “before” SWR DA peak, and one “after” SWR DA peak; the *BeforeAfter* categorical variable labels these data points as before or after an SWR. As a result, the regression can determine how
 495 much model fits are improved by adding this *BeforeAfter* information.

Following this initial analysis, we only included peak values after SWRs. To assess whether SWR-DA transients were experience-dependent, we ran session-based mixed effects regressions of the following forms:

$$\text{SWRDA} \sim 1 + (1 \mid \text{Mouse}) + (1 \mid \text{Session})$$

500
$$\text{SWRDA} \sim 1 + \text{PrePost} + (1 \mid \text{Mouse}) + (1 \mid \text{Session})$$

$$\text{SWRDA} \sim 1 + \text{EarlyLate} + (1 \mid \text{Mouse}) + (1 \mid \text{Session})$$

where *SWR-DA* denotes the peak of the session-averaged SWR-DA signal within one second after a SWR. *PrePost* indicates whether the session was a pre-task session or post-task session. *EarlyLate* indicates the first half of sessions (sessions 1-4) or later half of sessions (session 5-
 505 8).

To assess whether there was a relationship between DA-SWR strength and (1) the strength of putative DA reward prediction errors (measured during the task; Figure 1d), or (2) motivation, as measured by average running speed on the track, we ran session-based mixed-effects regressions of the following forms:

510
$$\text{SWRDAz} \sim 1 + (1 \mid \text{Mouse}) + (1 \mid \text{Session})$$

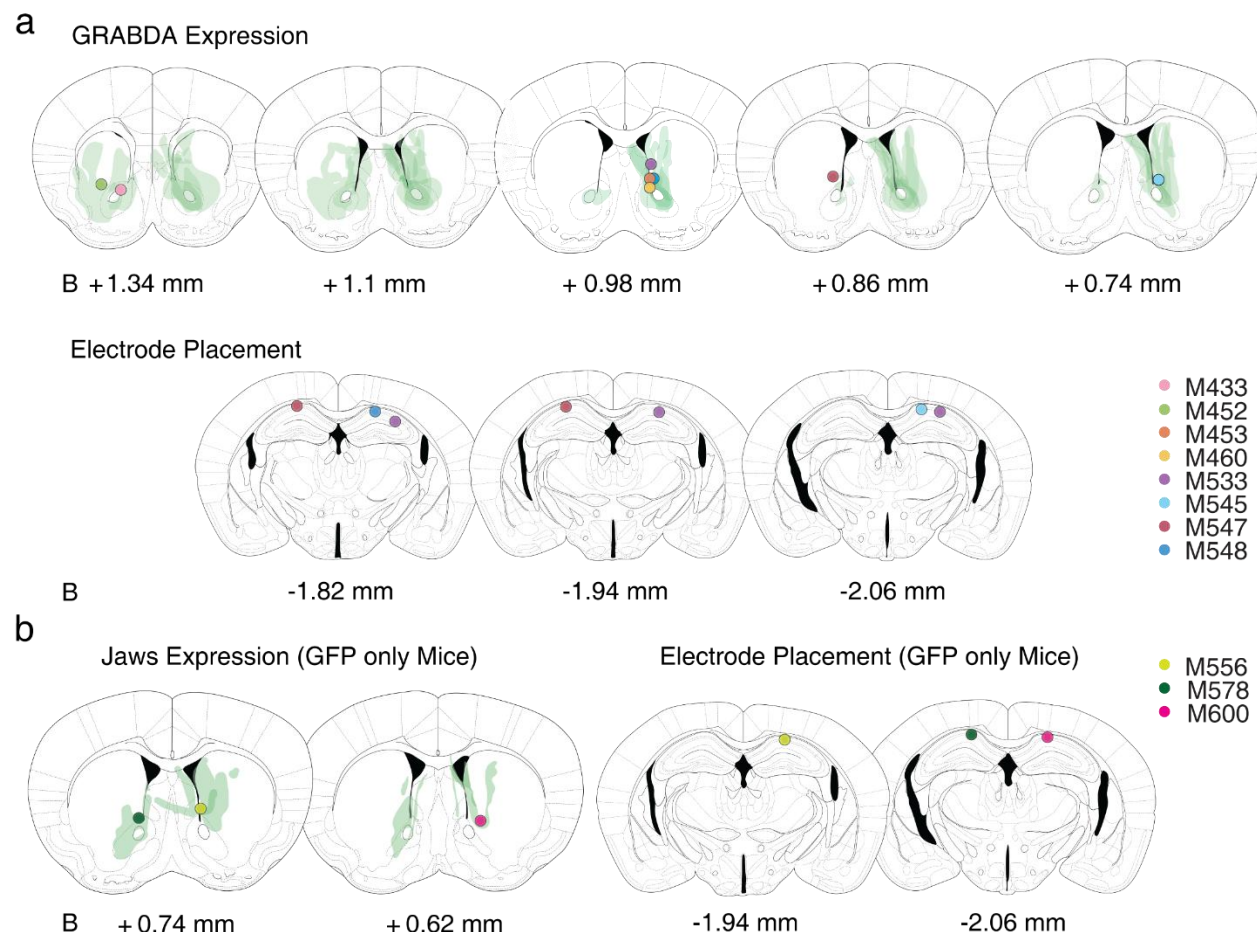
$$\text{SWRDAz} \sim 1 + \text{Motivation} + (1 \mid \text{Mouse}) + (1 \mid \text{Session})$$

$$\text{SWRDAz} \sim 1 + \text{Value} + (1 \mid \text{Mouse}) + (1 \mid \text{Session})$$

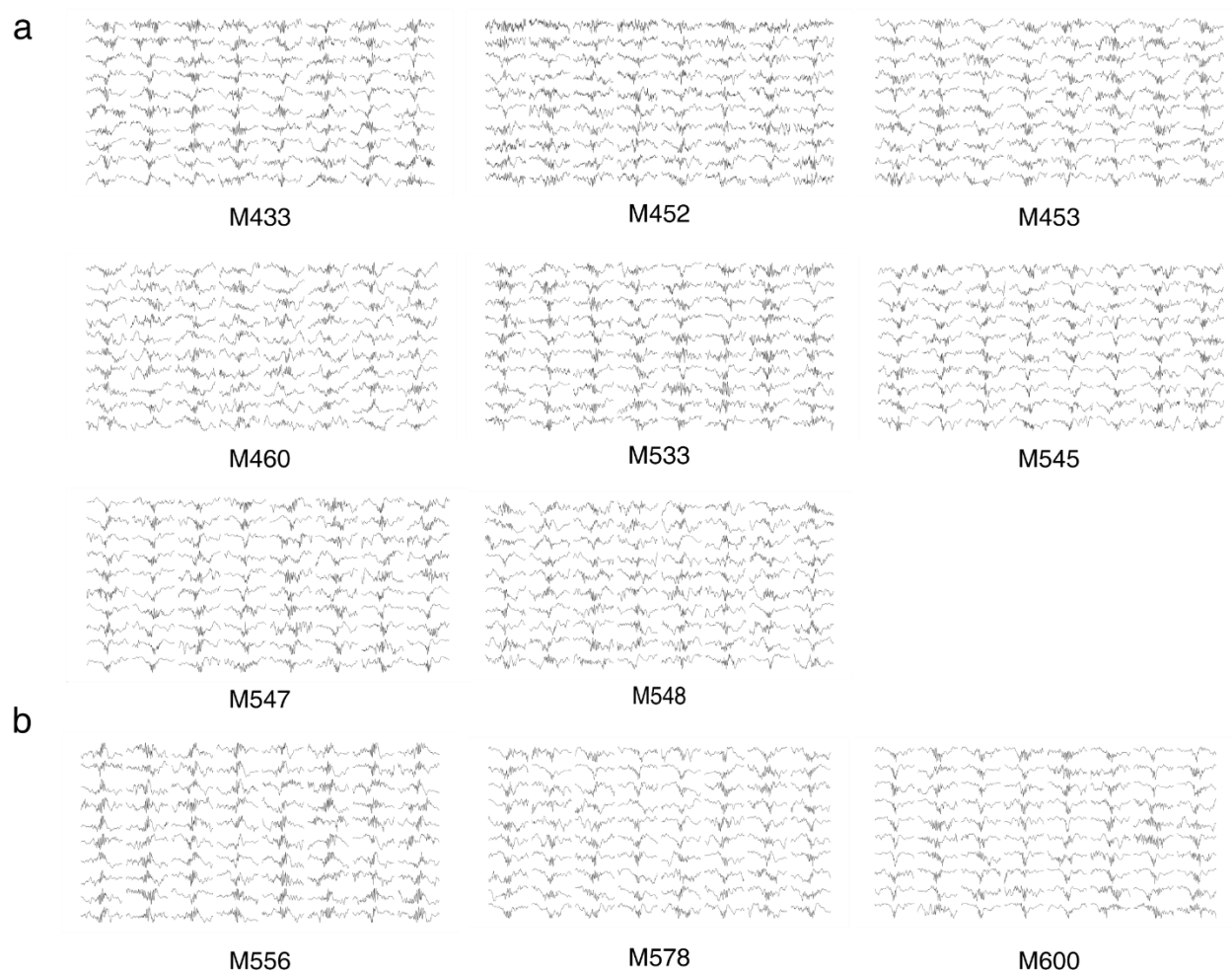
where *SWRDAz* denotes the permutation z-score of the SWR-DA coupling as described in the previous section (i.e. raw SWR-DA compared to a circular shuffle for each session). *Motivation*
 515 was defined as the average track speed (pixels/s). *Value* denotes the t-statistic between the peak [DA] signal for high reward trials and the trough for omission trials within an eight second period after reward zone entry.

Quantification and statistical analyses. Unless otherwise specified, we treated individual mice as the unit of analysis, and plots of aggregated data show mean $\pm$ standard error of the mean (SEM) over mice. Only sessions in which fiber recording locations were histologically verified to be within the ventral striatum, had power spectral density plots free from noise, raw fiber photometry amplitudes > 0.01 V, at least 100 detected SWRs, sufficiently motivated mice, and no recording failures were included in the analysis (N = 46 sessions across the 8 mice included for analysis, with a minimum of 4 sessions per mouse. We had to exclude mice with fiber locations outside of the ventral striatum (1 mouse) or if GRAB_{DA2m} was not well expressed (2 mice; none of these were included in the 8 mice analyzed here). Specifically, we excluded sessions due to recording failures, such as an optical fiber becoming unplugged (1 session). Additionally, sessions which had large noise in the power spectral density plots (6 sessions), in which raw fiber photometry amplitudes were below 0.01 V (1 session), or in which less than 100 sharp-wave ripples were detected (3 sessions) were excluded. Sessions in which the animal did not complete 60 trials in under 30 minutes were also excluded (1 session).

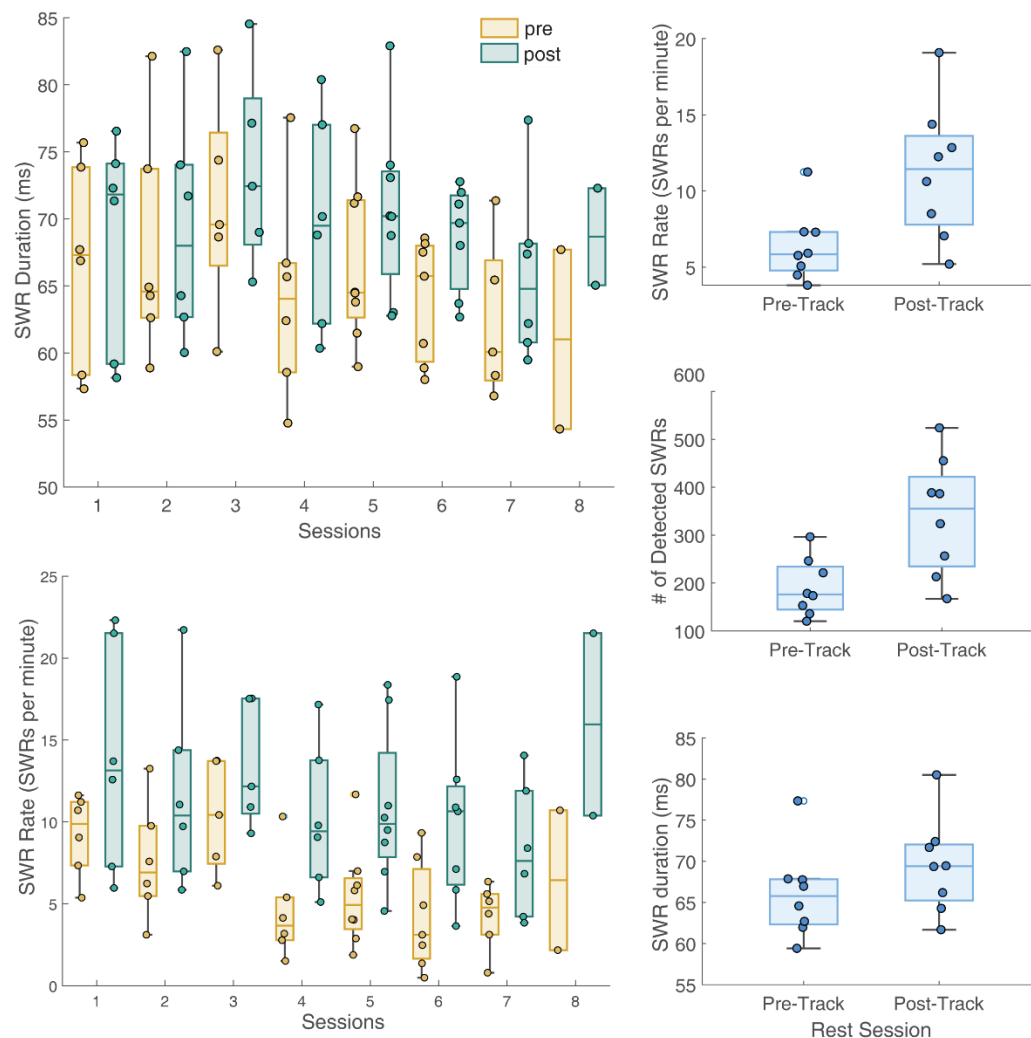
Supplemental Figures



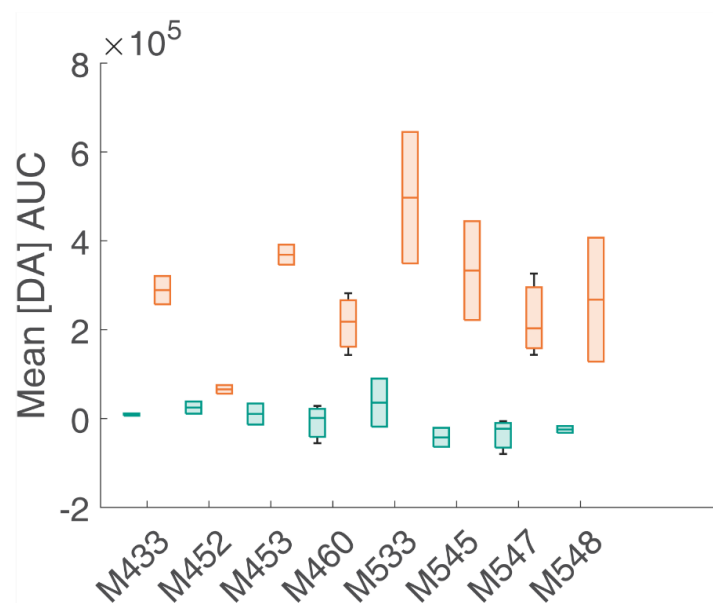
Extended Figure 1 | Histological verification of GRAB_{DA2m} expression, Jaws expression, and microdrive placement. **a**, *Top*: GRAB_{DA2m} expression maps. Each coronal section represents a section relative to Bregma (B, in mm). Expression for each animal is plotted at 20% transparency with per-animal expression overlaid. The estimated widest part of the fiber cannula is depicted by a filled circle marker with the color representing a different mouse. *Bottom*: estimated microelectrode placement. Each coronal section represents a section relative to Bregma (B, in mm). The estimated point of microelectrode placement is depicted by a filled circle marker (same as above) determined by post lesion gliosis. Three animals did not have clear gliosis, but did have neurophysiological signatures indicating dCA1 placement (extended figures 2&3) as well as clear microdrive cannula placement in the cortex above dCA1. **b**, as in **a**, for GFP only mice. Expression for each animal is plotted at 33% transparency with per-animal expression overlaid.



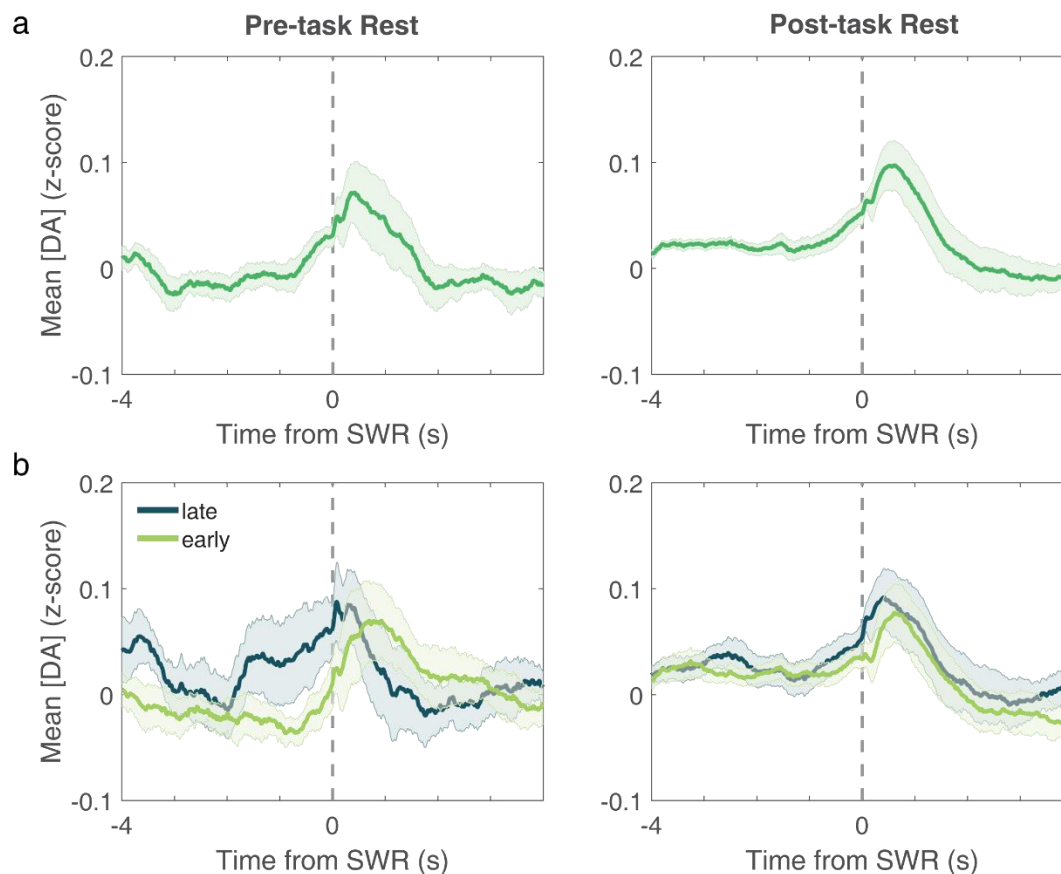
Extended Figure 2 | Example SWR traces from each mouse. a, 80 example detected SWRs from a session for each GRAB-DA mouse. **b**, 80 example detected SWRs from a session for each GFP-only mouse. We consistently observed clear SWR events in each included mouse, with no visible differences between GRAB-DA and GFP-only mice.



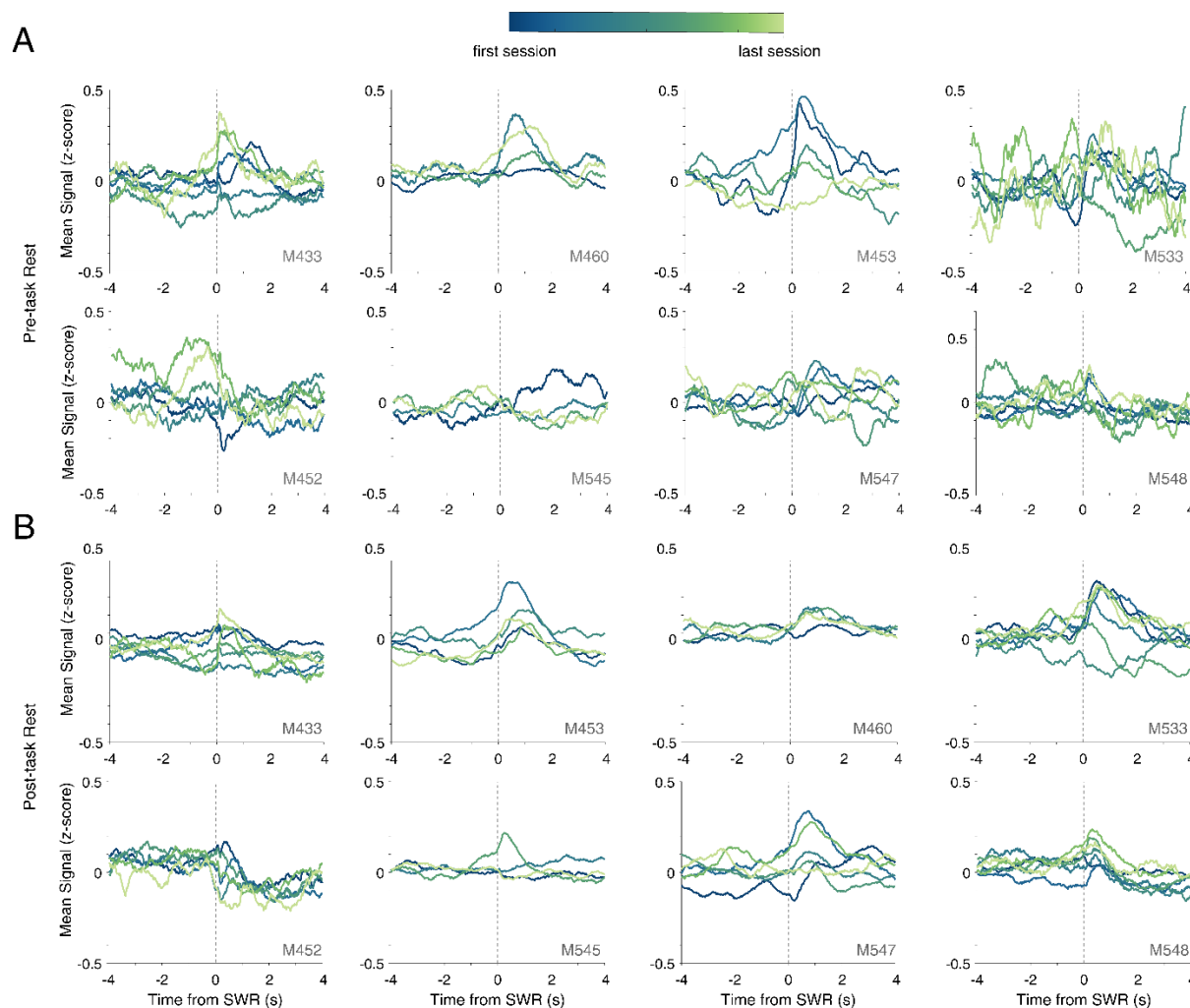
Extended Figure 3 | Sharp wave-ripple descriptive statistics. a, Top: SWR rate, defined as the average number of SWRs per minute. Yellow boxplots represent pre-task rest sessions and green boxplots represent post-task rest sessions. Each marker represents the average for a mouse for that session. The whiskers are the maximum and minimum. **Bottom:** Average length of SWRs in milliseconds (colors as above). **b, Top:** Average SWR count. The average number of detected SWRs, separated by pre- and post-task rest sessions. Each marker represents an average over sessions for a mouse. The whiskers are the maximum and minimum. **Middle:** Average SWR rate. The average number of detected SWRs over one minute, separated by pre- and post-task rest sessions. Each marker represents an average over sessions for a mouse. The whiskers are the maximum and minimum. **Bottom:** Average SWR duration. The average length of detected SWRs in milliseconds, separated by pre- and post-task rest sessions. Each marker represents an average over sessions for a mouse. The whiskers are the maximum and minimum.



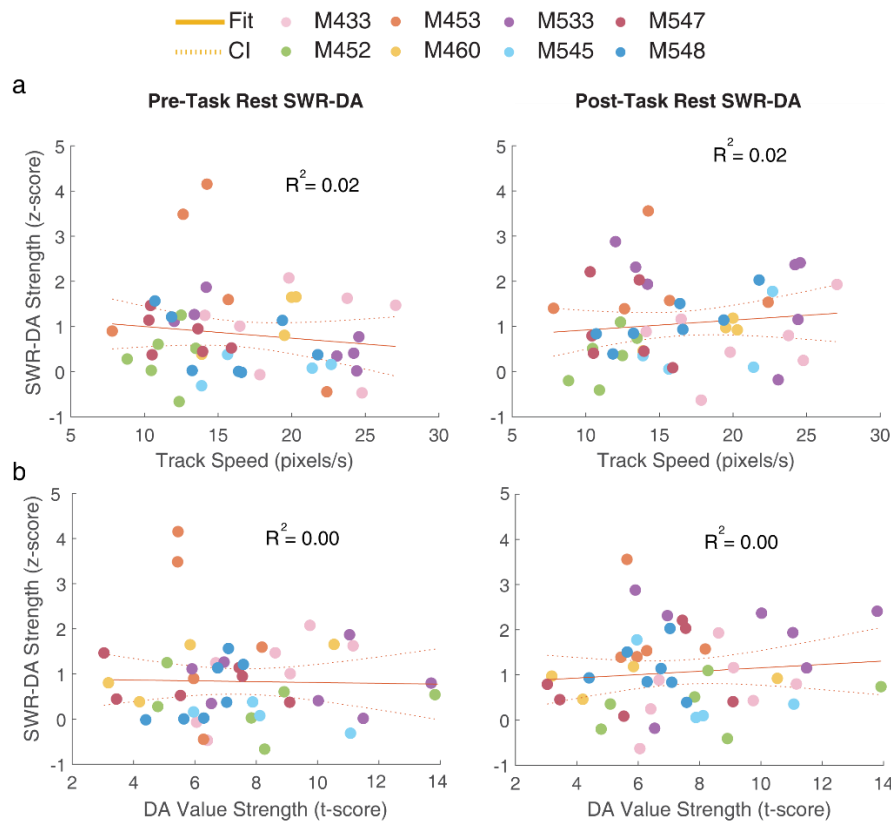
Extended Figure 4 | DA increase after amphetamine injection is seen consistently across mice. mean area under the curve of [DA] fiber trace during the incubation period subtracted by the instantaneous [DA] value at injection for experimental mice. Teal indicates saline sessions and orange indicates amphetamine sessions.



Extended Figure 5 | Peri-event time histograms for SWR-triggered dopamine response during the pre- and post-task epochs. a, Average [DA] response following SWRs split into pre- and post-task rest sessions. Shaded region indicates $\pm$ SEM ($n = 8$). **b,** Average [DA] response following SWRs split into early (sessions 1-4; 23 sessions total) and late sessions (sessions 5-8; 20 sessions total). Shaded region indicates $\pm$ SEM ($n = 8$).



Extended Figure 6 | Peri-event time histograms for SWR-triggered dopamine response for each individual session. a, Eight seconds of average DA response (z-score) centered on SWRs for each session for pre-task rest sessions. Each plot shows the average of each session for an experimental mouse. Sessions are graded in color with the first session in dark blue and the last session for that mouse in light green. **b**, as in **a**, for post-task rest sessions.



Extended Figure 7 | Linear regressions between SWR-DA Strength and either Track Speed or Teaching Signal Strength. **a**, Scatterplot comparing average track speed (pixels/s) with SWR-DA strength, the average peak in the [DA] trace within a one second period following SWRs (normalized relative to a control distribution of circularly shuffled SWR times; see Methods). Left plots show data from the pre-task rest session and right plots show data from the post-task rest session. Each point represents a session with color indicating mouse ID. A linear regression line is overlaid, demonstrating the overall trend and error bands around the regression line represent the 95% confidence interval. A LMM with random mouse-specific and session-specific intercepts and SWR strength as the dependent variable was not improved by the addition of track speed as a proxy for motivation for either the pre- or post-task rest sessions. (pre: likelihood ratio test, $\text{LRStat}_{\{\Delta d.f. = -1\}} = 1.73$, $p = 0.21$. post: likelihood ratio test, $\text{LRStat}_{\{\Delta d.f. = -1\}} = 0.41$, $p = 0.57$). **b**, Scatter plot comparing SWR-DA strength with teaching signal strength, which is the t-value between the peak [DA] signal response for high volume trials and the trough for omission trials within an eight second period after reward zone entry. Each point represents a session with color indicating mouse ID. A linear regression line is overlaid, demonstrating the overall trend and error bands around the regression line represent the 95% confidence interval. A LMM with random mouse-specific and session-specific intercepts and SWR strength as the dependent variable was not improved by the addition of value strength for either the pre- or post-task rest sessions (pre: likelihood ratio test, $\text{LRStat}_{\{\Delta d.f. = -1\}} = 0.03$, $p = 0.89$. post: likelihood ratio test, $\text{LRStat}_{\{\Delta d.f. = -1\}} = 0.67$, $p = 0.45$).